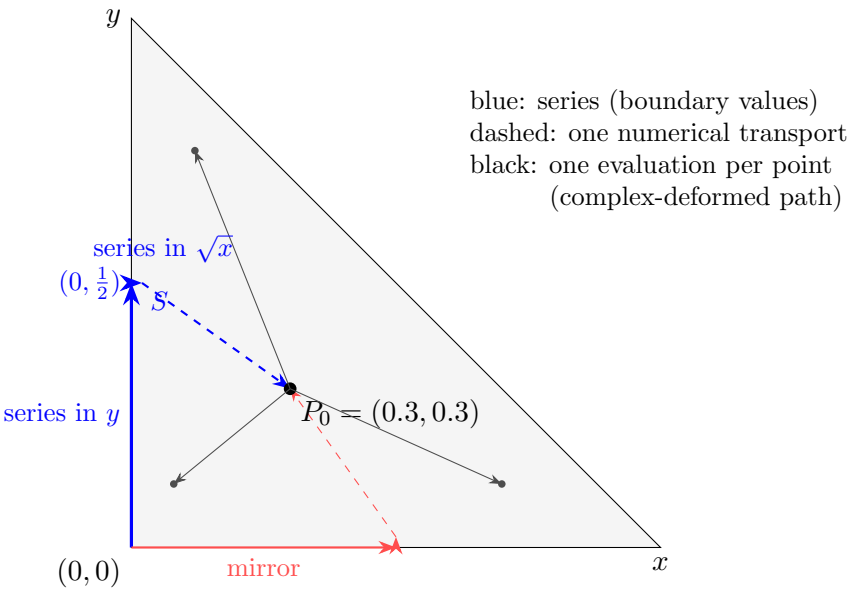
Numerical evaluation of the $V \to q\bar{q}g$ finite remainders: boundary values and transport

Setup. Every finite remainder is a sum $\sum q\, R_r(y,z)\, T_g\, c_m$ of rational functions, constants and 579 pure functions T_g. The T_g are iterated integrals with base point $(x,y) = (0,0)$, where every non-empty integral vanishes after shuffle regularisation. They satisfy a closed canonical system $dF = \sum_k d\log W_k\, M_k F$ of 503 functions, which we integrate numerically.

Boundary values (computed once). The origin is singular, so we do not start there. We expand in series instead (see the figure):

1. along the edge $x = 0$, from $(0,0)$ to $(0, \frac{1}{2})$, as a series in y. On the edge only $y = 0, 1$ are singular, and the regularisation sets $\log y \to 0$ at the origin;
2. off the edge at fixed $y = \frac{1}{2}$, as a series in $\sqrt{x}$ (the square-root letters go like $\sqrt{x}$), up to $S = (0.02, \frac{1}{2})$. Here $\log x \to 0$ at $x = 0$;
3. one numerical transport (Bulirsch–Stoer) from S to the central point $P_0 = (0.3, 0.3)$.

The values at P_0 are stored once and for all and shipped with the evaluator. The mirror construction, through $(\frac{1}{2}, 0)$ and $(\frac{1}{2}, 0.02)$, gives an independent set, used only for the error estimate.



Evaluation at a point P. We integrate the system from P_0 to P along a straight path, deformed into the complex plane ($z(t) = P_0 + (t + 4i\delta\, t(1-t))(P - P_0)$). Three refinements:

- *Spurious letters.* Six letters vanish on curves inside the triangle. The amplitude is smooth there, but the equations are not. The deformation δ is chosen so that the complex path stays away from their zeros.

- *Points on a spurious curve.* Such a point is replaced by the symmetric average of $P \pm h\,n$, $P \pm 2h\,n$, $P \pm 3h\,n$ across the curve (n its normal), which is accurate to $O(h^6)$.

- *Error estimate.* Each point is evaluated a second time, from the mirror boundary values and on another path; the estimate is ten times the difference.

In the bulk the accuracy is about 10^{-12}, and about 10^{-13} against the exact planar results. Near the edges double precision loses digits, which the error estimate reports; there the corner expansions take over.